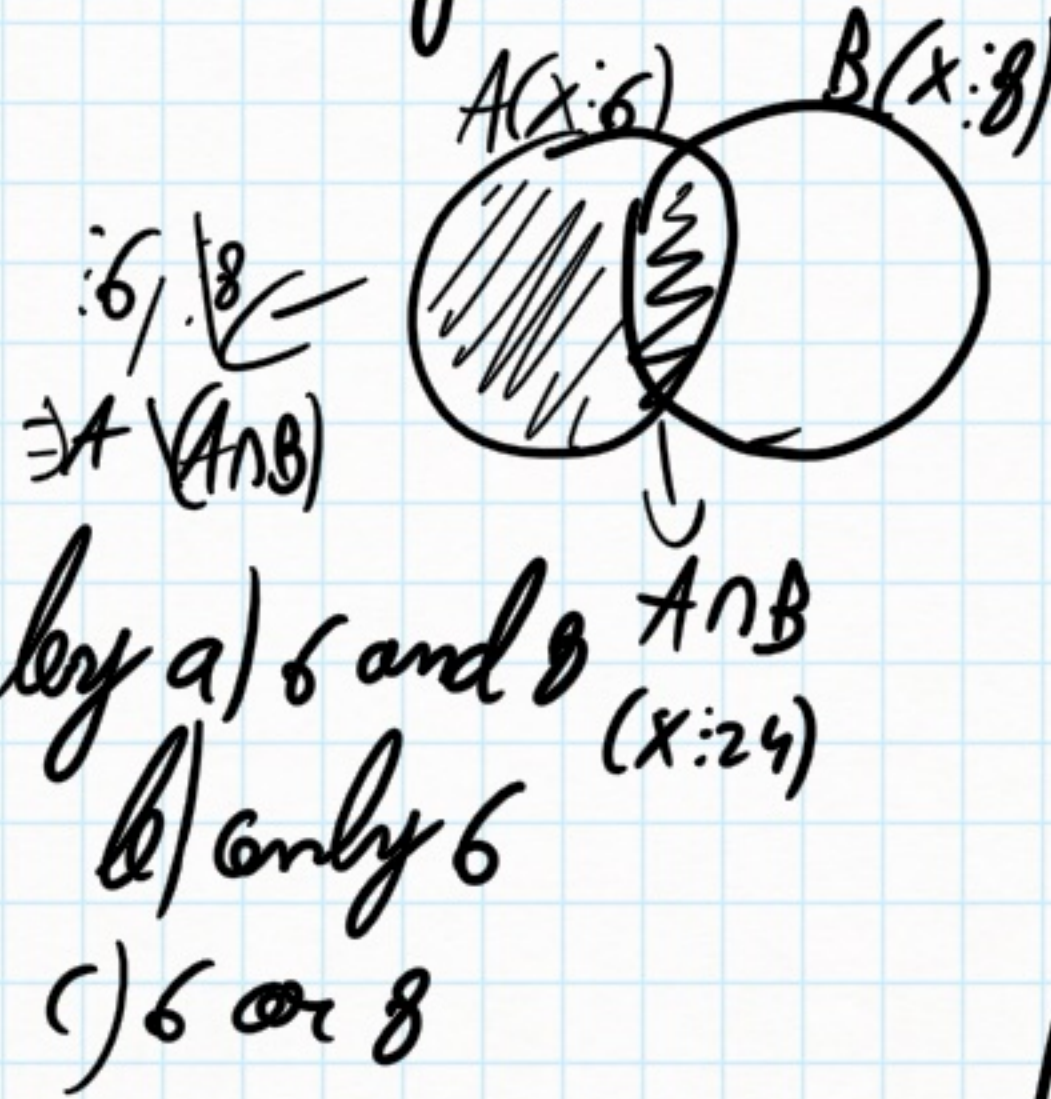


# Probability and Statistics

## Seminar 1

### Counting Problems

- ①. Consider the set  $A = \{1, 2, \dots, 200\}$   
 how many elements of  $A$  are divisible by a) 6 and 8  
 b) only 6  
 c) 6 or 8



!  $M \cap N = \emptyset \Rightarrow$  disjoint

a)  $6 = 1 \cdot 2 \cdot 3$   
 $8 = 1 \cdot 2 \cdot 4$

$24 \cdot 8 = 192$

a)  $M_{24} = \{24, \dots, 8 \cdot 24\}$   
 $\Rightarrow 8$  elem.

b)  $\text{Card}(A \setminus (A \cap B)) = \text{card}(A) - \text{card}(A \cap B)$   
 $= 33 - 8 = 25$

Seminar: 2 tests + activity  
 + essay again

option 1  
 2 pts  
 something in IT  
 where u apply  
 PS

option 2  
 3 pts  
 needs extra  
 research

there is OPTIONAL MIDTERM  
 1 hr/part

exam: up to 2 hrs  
 at least half of exam passed = passed seminar

$$c) \text{card}(A \cup B) \begin{cases} \text{card}(A) + \text{card}(B) & \text{IF DISJOINT} \\ \text{card}(A) + \text{card}(B) - \text{card}(A \cap B) & \text{otherwise} \end{cases}$$

$\begin{matrix} // & // & // \\ 33 & 25 & 8 \end{matrix}$

$$C_m^h = \frac{m!}{h!(m-h)!} = \binom{m}{h} \text{ "m choose h"}$$

$$\text{card}(A \cup B \cup C) = \text{card} A + \text{card} B + \text{card} C - \text{card}(A \cap B) - \text{card}(A \cap C) - \text{card}(B \cap C) + \text{card}(A \cap B \cap C)$$

② Prob. ex  $\rightarrow$  2 parts  $\begin{cases} \text{I: 5 problems} \\ \text{II: 4 problems} \end{cases}$

Each student has to solve 5 problems, at least 2 from part I and at least 2 from part II

5 plb:  $\begin{cases} 3I+2I \\ 2I+3II \end{cases}$

$$\binom{3}{5} \binom{2}{4} + \binom{2}{5} \binom{3}{4} = \frac{5!}{2! \cdot 3!} \cdot \frac{4!}{2! \cdot 2!} + \frac{5!}{2! \cdot 3!} \cdot \frac{4!}{3!} \\ = 10 \cdot 6 + 10 \cdot 4 = 100$$

$f: A \rightarrow B$

nr. of functions:  $(\text{card } B)^{\text{card } A}$

Nr of subsets of  $A$ :  $2^{(\text{card } A)}$

Nr of subsets with  $m$  elements:  $\binom{n}{m}$   $\rightarrow m \leq \text{card } A$

$$\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$$

③. 8 W 7 M  
Mrs. X, Mr. Y

committee of 3 W, 4 M

Mrs. X refuses to participate with Mr. Y

$$\binom{3}{8} \cdot \binom{4}{7} - \binom{2}{7} \cdot \binom{3}{6} = 1960 - 420 \\ = 1540$$

④. How many functions  $f: \{1, 2, \dots, 10\} \rightarrow \{0, 1\}$   
 have the property:  $f(1) + f(2) + \dots + f(10) = 2$   
 $1 + 2 + 3 + \dots + 9 = 9 \cdot 9 \cdot 4 = 9 \cdot 5 = 45$

Arrangements *yes* combinations  
 counts  $\nwarrow$  order  $\nearrow$  doesn't count

⑤. Nr. of surjective functions  $f: \{1, 2, 3\} \rightarrow \{1, 5\}$   
 Total nr. of functions - nr. of  $f$  not surjective  
 $2^3 = 8$  —  $f(1) = f(2) = f(3) = 1$   
 or  $f(1) = f(2) = f(3) = 5$   
 $2 = 6$

6. Nr. of even functions  $f: A \rightarrow R$   
 $A = \{-2, -1, 0, 1, 2\}$

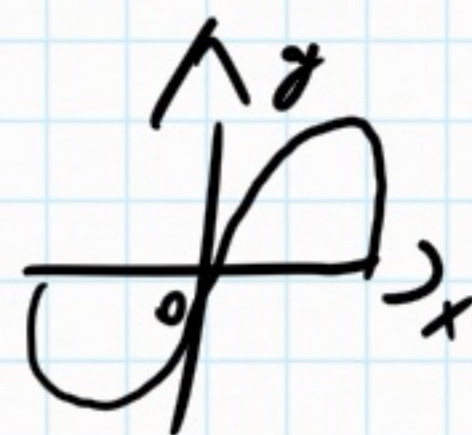
even:  $f(-x) = f(x), \forall x \in D$

odd:  $-f(x) = f(x), \forall x \in D \rightarrow x=0 \Rightarrow f(0) = -f(0) \Rightarrow f(0)$

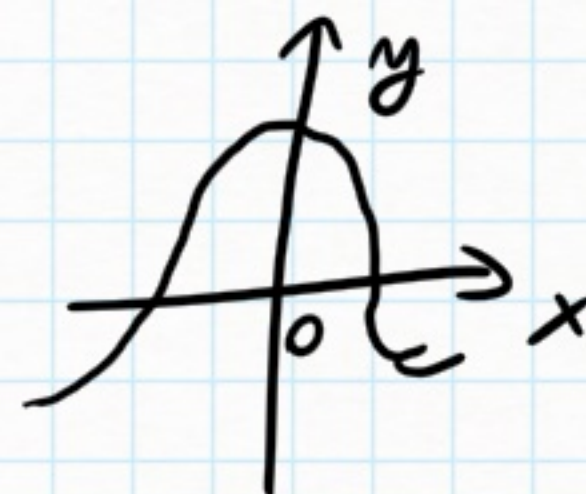
$f: \underset{-1 \quad -2}{\{0, 1, 2\}} \rightarrow \{-2, -1, 0, 1, 2\}$

$$R: 5^3 = 125$$

$f(x) = \sin x$   
odd



$f(x) = \cos x$   
even



7 Nr. of odd functions

$f(0) = 0, f(-1) = -f(1)$

$f(-2) = -f(2)$

$\{1, 2\} \rightarrow \{-2, -1, 0, 1, 2\} \Rightarrow R: 5^2 = 25$

8. How many binary strings of length 8 start with 1 and end with 00?

$$\begin{array}{c} 1 \text{ --- } 00 \\ 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \\ = 2^5 = 32 \end{array}$$

9. -||- start with 1 or end with 00

$$2^7 + 2^6 - 2^5 = 128 + 64 - 32 = 160$$

A: 1 - - - - -

B: - - - - - 00

$$\text{card}(A \cup B) = \text{card } A + \text{card } B - \text{card}(A \cap B)$$

9. How many subsets of  $\{1, 2, \dots, 10\}$  has the product of the elements: 10?

1 2 3 4 5 6 7 8 9 10

I. all which contain 10:  $2^9$

II. all without 10:

5 1 {2, 4, 6, 8} 1 {1, 3, 7, 9}

↓

↓

↓

1 ·  $(2^4 - 1)$

$2^4$

$= 16 \cdot 15 = 240$

Total: 752 out of  $2^{10}$   
subsets